OPPORTUNITY BRIEF · SALES AND PRESALES

Digital product passports for construction materials to ensure circularity and compliance with EU ESPR regulations, leveraging blockchain for tamper-proof traceability.

Construction × Product traceability and digital product passport × Blockchain and DLT · OS050 v2 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
62 is the world moving	**13** can we play, can we win	**LOW** 1 named in the evidence	**€4.7bn** partial evidence · per year	**NEXT** regulation adopted or propose...	**L3** 15 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 1.0 published reference(s) in Construction, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity is about helping construction material manufacturers and their supply chains prepare for mandatory Digital Product Passports (DPPs) under the EU's Ecodesign for Sustainable Products Regulation (ESPR). With European technical standards now in place and the regulation approaching applicability, there is a clear window for Orange Business to deliver a blockchain-based traceability solution that ensures compliance and supports circularity. The opportunity exists now because the regulatory framework is defined but not yet enforced, giving early movers a competitive advantage.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	**€10.4bn** €545m – €50.5bn	**€4.7bn** €244m – €22.6bn	**€5.8m** €305k – €28.3m	partial
Observed: tendered contract value, annualised	**€123m** €123m – €471m	**€123m** €123m – €471m	**€154k** €154k – €588k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Construction (EU27_2020)	250,045 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Blockchain and DLT	48.5 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cicce_usen2 · E_CC	2025
Annual value of one engagement	€952k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	22,551 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Cloud computing services by NACE Rev. 2 activity series E_CC is used as a proxy for Blockchain and DLT: No DLT series at NACE level. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

Observed: tendered contract value, annualised

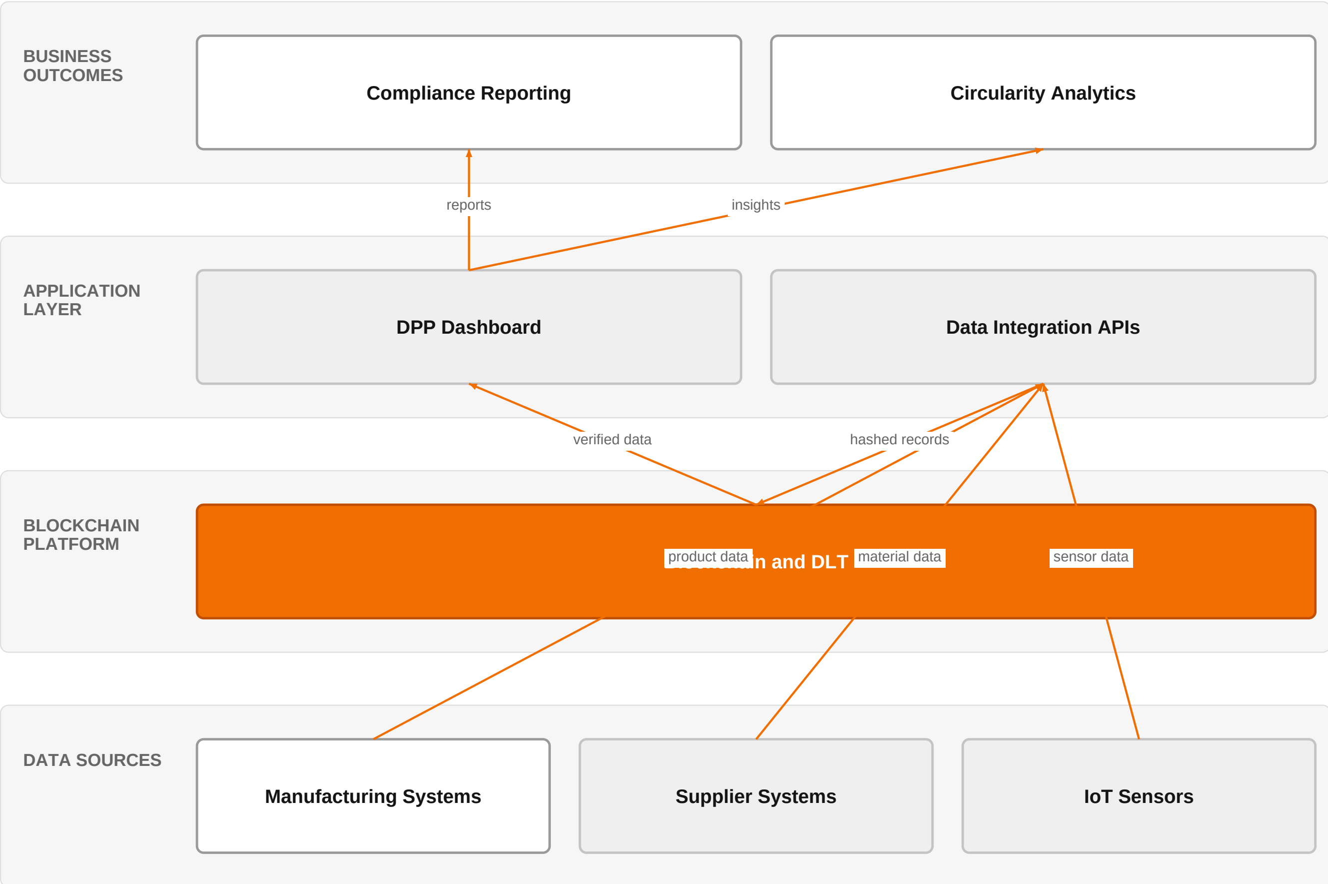
Factor	Value	Basis	Source	As of

Tendered contract value in matching CPV categories	€123m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 24 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Blockchain-based DPP for Construction Materials



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how Orange's blockchain and DLT platform securely aggregates data from manufacturing, suppliers, and IoT sensors to power compliance reporting and circularity analytics.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The EU's Ecodesign for Sustainable Products Regulation (ESPR) mandates Digital Product Passports for a wide range of product categories, creating an urgent need for machine-readable product lifecycle data. European technical standards for DPPs have now been published, clarifying the operational requirements. The construction sector, being one of the most resource-intensive, is under pressure to adopt circularity practices, and the BUILDERS project is testing an Extended Producer Responsibility scheme for construction products. Research highlights significant knowledge gaps in the construction industry regarding DPP implementation, indicating a need for guidance and solutions.

`Sources: SIG-AA85D6CD0C67 openalex.org 2026-02-27; SIG-AF45FABC720F RECYCLING magazin 2026-08-03; SIG-36E0C3234EE0 cordis.europa.eu 2026-06-01; SIG-EFEBD520406D openalex.org 2025-10-15`

Who buys, and why now

The primary buyer is the Chief Sustainability Officer or Head of Compliance at a construction materials manufacturer, who feels the pain of upcoming regulatory deadlines and the complexity of tracing materials across fragmented supply chains. The trigger is the publication of technical standards and the approaching ESPR applicability, which forces them to act to avoid market access barriers. Operational pain includes manual data collection, lack of interoperability with suppliers, and inability to demonstrate circularity to customers and regulators.

Why it is hot now — every claim bound to a dated source

- The BUILDERS project addresses the urgent need for circularity in Europe's construction sector through an Extended Producer Responsibility scheme. [SIG-36E0C3234EE0]
- DigiPass aims to enhance digital maturity of European materials communities and pave the way to a Digital Materials & Product Passport. [SIG-1CB3CD955DD7]
- The BUILDERS project addresses the urgent need for circularity in Europe's construction sector, one of the most resource-intensive. [SIG-36E0C3234EE0]
- Digital Product Logbook is proposed as a repository for logging data after the product is placed on the EU Single Market. [SIG-8E9F663B741B]
- The BUILDERS project addresses the urgent need for circularity in Europe's construction sector through an Extended Producer Responsibility Scheme for construction products. [SIG-36E0C3234EE0]
- The Digital Product Logbook is proposed as a repository for logging data after the product has been placed on the EU Single Market, applicable to the built environment. [SIG-8E9F663B741B]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its Blockchain and DLT technology, combined with its AI/Data/Cloud experts and Digital experts. The engagement would start with a discovery workshop to map the customer's material flows and data sources, followed by designing a DPP data model aligned with EU standards. Then, Orange would implement a blockchain-based traceability platform that records product lifecycle data immutably, integrating with existing ERP systems via APIs. Finally, Orange would provide a dashboard for compliance reporting and circularity analytics, leveraging its cloud and data capabilities.

Why Orange can win

Orange can win because it combines blockchain and DLT expertise with deep data and cloud capabilities, enabling a secure, scalable DPP solution. Orange's position as a trusted operator, with a focus on sovereignty, aligns with the regulatory and trust requirements of DPPs. However, the assets are thin in terms of specific construction domain expertise, so Orange must partner or build that knowledge to compete effectively.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Blockchain and DLT	technology	L3	no offer path, but adjacent assets or Orange research exist	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Healthcare experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

LOW competitive intensity (score 1.8; 1 of 8 listed players appear in this topic's own evidence)

One or two adjacent players. Orange would be shaping the space rather than fighting for it.

Competitors include hyperscalers like AWS, Google Cloud, and Microsoft, who offer cloud platforms that can host DPP solutions, and systems integrators like Accenture and Atos/Eviden who can build custom implementations. Databricks, named in the evidence, provides data and AI platforms that could support DPP data management. Industrial specialists like Schneider Electric and Siemens have domain expertise in construction and smart industries, making them strong in OT integration. The customer will compare Orange's end-to-end managed service and sovereign credentials against these alternatives.

Sources: SIG-BC952EFA76FC Databricks 2026-07-27

Competitor	Category	Position here	Basis	Evidence
Databricks	Data and AI platform	competes in Cloud	evidenced	SIG-BC952EFA76FC
Schneider Electric	Industrial / OT specialist	active in Construction; competes in OX: Smart Industries	structural	—
Siemens	Industrial / OT specialist	active in Construction; competes in OX: Smart Industries	structural	—
AWS	Hyperscaler	competes in Cloud, OX: Smart Industries — also an Orange partner	structural	—
Google Cloud	Hyperscaler	competes in Cloud — also an Orange partner	structural	—
Microsoft	Hyperscaler	competes in Cloud — also an Orange partner	structural	—
Accenture	Global systems integrator	competes in Cloud, OX: Smart Industries	structural	—
Atos / Eviden	Global systems integrator	competes in Cloud	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

1. Are you currently preparing for the EU Digital Product Passport requirements under ESPR, and what is your timeline for compliance?
2. How do you currently track the lifecycle of your construction materials from production to end-of-life, and what data gaps exist?
3. Have you engaged with any of the EU-funded projects like BUILDERS or DigiPass to pilot DPP solutions?
4. What is your biggest challenge in achieving circularity for your products: data collection, supplier collaboration, or regulatory reporting?
5. Do you have a preference for a sovereign or cloud-based solution, and what are your security requirements for product data?

Objections you will meet

They will say	You can answer
We already have an ERP system and don't see the need for a separate blockchain solution.	That's a fair point, and your ERP is likely the source of truth for transactional data. However, DPPs require tamper-proof, lifecycle-wide traceability that spans multiple organizations, which traditional ERPs are not designed to provide. Our blockchain layer can complement your ERP by creating an immutable record that is shareable with regulators and partners.
The ESPR requirements for construction products are not final yet, so why should we invest now?	You're right that the final delegated acts are pending, but the technical standards are already in place, and the direction is clear. Early movers will have a competitive advantage in shaping their data collection processes and avoiding costly retrofits. Our solution is designed to be flexible and can adapt to final requirements.
We are a small manufacturer and cannot afford a complex IT project.	We understand budget constraints. Our approach is modular, allowing you to start with a minimal viable DPP that meets core compliance needs, and scale as requirements evolve. We also offer managed services to reduce the burden on your IT team.

Risks and unknowns

The main risk is that the ESPR applicability for construction products is not yet fully defined, and the final data requirements may shift. There is also uncertainty about how DPPs will interoperate across the value chain, as semantic web vocabularies are still evolving. Additionally, the construction industry's low digital maturity may slow adoption, and the cost of implementation could be a barrier for smaller players. Orange must also prove its ability to integrate with legacy systems and handle the scale of construction data.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a European construction materials association, outlining how Orange's Blockchain and DLT technology, combined with our AI/Data/Cloud and IoT expertise, can deliver a tamper-proof digital product passport solution that meets ESPR circularity goals.
Sales	In a conversation with a European construction materials manufacturer's sustainability director, say: 'We have deep expertise in AI, data, and IoT that can help you prepare for the EU's upcoming digital product passport requirements, ensuring your materials are compliant and traceable.'
Strategist / Innovator	Commission a deep-dive brief on the BUILDERS and DigiPass projects to map how Orange's AI/Data/Cloud and IoT experts could support a Digital Product Passport pilot for construction materials, focusing on the ESPR compliance timeline.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-36E0C3234EE0	2026-06-01	cordis.europa.eu	1	Building Up Industry Loops through Development, evaluation, and testing of an Extended producer...
SIG-36FFB2435F8A	2026-04-20	openalex.org	1	A Comprehensive Review of Digital Product Passports in Sustainable Manufacturing: Current State...
SIG-8E9F663B741B	2026-03-30	Informes de la Construcci...	1	Digital Product Logbook, the missing piece in the European Digital Product Passport puzzle
SIG-AA85D6CD0C67	2026-02-27	openalex.org	1	Digital product passports
SIG-F47C31AB60CB	2026-02-14	MDPI AG	1	Dynamic Traceability and Reliability Management: Integrating the Digital Product Passport with ...
SIG-BA196FF05244	2026-02-03	Springer Science and Busi...	1	Digital Product Passport for Circular Economy: A Review of Semantic Web Vocabularies and Requir...
SIG-4FE21FD2ED06	2026-01-01	openalex.org	1	On the Utilization of Digital Product Passports for Intelligent Metal Sorting and Recycling
SIG-CC28A63D0BB7	2026-01-01	openalex.org	1	From Compliance to Circularity: Stakeholder Perspectives on Adopting Digital Product Passports ...
SIG-EFEBD520406D	2025-10-15	openalex.org	1	Stakeholder perspectives on digital product passports for construction products
SIG-40DC35D6534F	2025-10-14	openalex.org	1	Predictive Generation of Digital Product Passport as Decision Support in Circular Networks
SIG-6FBECA0F7EA7	2025-09-10	openalex.org	1	Design of a Sensor-Based Digital Product Passport for Low-Tech Manufacturing: Traceability and ...
SIG-1CB3CD955DD7	2024-04-01	cordis.europa.eu	1	Harmonization of Advanced Materials Ecosystems serving strategic Innovation Markets to pave the...
SIG-AF45FABC720F	2026-08-03	RECYCLING magazin	2	Digital Product Passport: European Technical Standards Now in Place - RECYCLING magazin
SIG-BC952EFA76FC	2026-07-27	Databricks	2	The EU Digital Product Passport: a traceability deadline - Databricks
SIG-A3AAC433E98C	2026-05-07	HKTDC Research	2	Proposed EU Rules Clarify Operation of Digital Product Passport Registry - HKTDC Research

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:43:27+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.